MUSTAFA HIJAZI

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EDUCATION

Politecnico di Milano | M.Sc. in Mechanical Engineering 2026 – Present
Automotive and Motorsport Engineering Track

American University of Beirut: B.E. in Mechanical Engineering 2022 - 2026

WORK HISTORY

Origami – Technology (Milano, Italy) | Hydrodynamics & Marine Mechanical Engineer
09/2026 – Present

- Currently designing the first prototype for a floating buoy that generates power via ocean wave movement.

Jubaili Bros (Saida, Lebanon) | Engineering Intern 08/2025

- Assisted in assembling power generators, control panels, and electrical layouts; reviewed chassis design and fabrication for generator housings.

Alef Group (Sharjah, UAE) | Mechanical Engineering Intern 06/2025 - 07/2025

- Reviewed HVAC, drainage, fire protection, gas, and water layouts for residential projects; drew HVAC layouts for villas using AutoCAD and conducted site inspections.

PROJECTS

Tri-copter with 360° Omnidirectional Maneuvering – Final Year Project 08/2025 – 11/2025

- Designed a custom UAV with a carbon fiber frame and 3D printed PETG components, focusing on strength and weight reduction; exposed the frame to 9G loading and verified its structural integrity

Aerial Manipulator – Aerial Robotics Course 08/2025 – 11/2025

- Assisted researching humanlike trajectories using minimum-jerk optimization; successfully simulated a smooth trajectory for a 6-DOF robotic arm on Gazebo using ROS2 and Python

Autonomous Firefighting UAV 09/2025 – 11/2025

- Developed logic for the scooping-phase control loop of an autonomous firefighting aircraft based on the CL-415; drafted the control architecture, block, and state diagrams for the UAV

SKILLS

- Software: SolidWorks, AutoCAD, Creo Parametric, MATLAB, Arduino, Ansys Fluent, MS Office
- Technical Skills: CAD modelling, Robotics, UAV design, Prototyping

COMMUNITY SERVICE/ Extracurriculars

Badiri NGO, Charity, Saida, Lebanon 09/2024 – 10/2024

Licenses and Certifications

- Data Fundamentals – IBM SkillsBuild 07/08/2026
- Data Classification – IBM SkillsBuild 03/09/2026